

Samuel B. Phillips

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EDUCATION

University of North Carolina at Charlotte

B.S. Computer Science (Data Science Track)

Charlotte, NC

Sep. 2025 – May 2028

University of North Carolina at Charlotte

B.S. Mathematics (Concentration in Statistics)

Charlotte, NC

Sep. 2023 – May 2027

- GPA: 3.69 (cumulative, both degrees) | Sigma Xi, Inductee

RESEARCH EXPERIENCE

Arctic PFAS Transport Modeling

UNC Charlotte — Dr. Roger Tipton (NSF Grant #2444939)

Charlotte, NC

Sep. 2025 – Present

- **Fall 2025 (OUR Scholar, Ocean Simulation Researcher):** Configured and executed global-domain MITgcm ocean simulations modeling passive microplastics transport; built Python preprocessing pipelines (NumPy, scikit-learn) integrating JRA55-do atmospheric forcing data. Presented at the UNC Charlotte OUR Honors Symposium.
- **Spring 2026 (OUR Scholar, Computational Simulation Researcher):** Designed and executed MITgcm LLC270 Arctic Ocean simulations modeling PFAS passive transport under varying atmospheric forcing conditions on HPC infrastructure. Presented at NCUR 2026 and the UNC Charlotte Undergraduate Research Conference (URC) 2026.
- **Summer 2026 (in progress):** Developing a physics-informed neural network (PINN) framework to correct TOPAZ/CMEMS Arctic Ocean simulation outputs against sparse observational PFAS data (~140 samples over 20 years); targeting AGU Fall 2026.

Representation Learning for Biogeochemical State Reconstruction

UNC Charlotte — Dr. Xuyang Li (OUR Scholar)

Charlotte, NC

Jan. 2026 – Present

- **Spring 2026:** Modeled latent ocean dynamics from Argo float profiles using Neural ODEs to reconstruct unseen ocean variables from learned trajectories in a compressed latent space; outperformed simple baselines (e.g., MLP).
- **Summer 2026 (in progress):** Developing a transformer-based architecture for biogeochemical state reconstruction from sparse Argo float observations, benchmarked against an existing convolutional approach; targeting *Geoscientific Model Development* (GMD).

Neural Time Series Modeling for Food Safety Prediction

NSF REU, Worcester Polytechnic Institute — Advisor: Dr. Fabricio Murai

Worcester, MA

May 2026 – Present

- Selected as 1 of 14 participants nationwide for the NSF REU in Applied AI; serving as Site Lead, liaising between the cohort and program directors.
- Developing and evaluating deep learning approaches — including time series foundation models and multimodal architectures — to predict and explain bacterial growth, contamination, and spoilage in food systems over time; targeting MIT's Undergraduate Research Conference.

Representation Geometry in Multi-Agent Reinforcement Learning

UNC Charlotte — with Jake Minwu Lee

Charlotte, NC

Summer 2026 – Present

- Surveying representation geometry in multi-agent reinforcement learning systems, building toward a proposal for UNC Charlotte's undergraduate research grant program (SURG).

INDUSTRY EXPERIENCE

AI/ML Engineer Intern

FanConnect

Charlotte, NC (Remote)

Jan. 2026 – Present

- Architected and shipped FanConnect Labs, a client-facing AI content generation platform (React/Node.js) with real-time preview and iterative refinement.
- Built an AI-assisted ad generation tool for FanConnect's marketing and social media campaigns.
- Developing a local vision-language model (VLM) to detect live game feed failures across FanConnect's venue network and trigger support alerts.

TEACHING

Teaching Assistant — ITCS 2214: Data Structures & Algorithms

Charlotte, NC

UNC Charlotte

Sep. 2025 – Dec. 2025

- Held office hours and evaluated student Java implementations of core data structures (hash tables, binary trees, graph algorithms).

PRESENTATIONS & TALKS

- **“Modeling the Transport of Forever Chemicals in the Arctic”** — National Conference on Undergraduate Research (NCUR), 2026
- **“Learning Ocean Dynamics from Sparse Data Using Scientific AI”** — UNC Charlotte Undergraduate Research Conference (URC), 2026
- **“Using Data Science to Model Global Microplastic Transport”** — UNC Charlotte OUR Honors Symposium, 2025

HONORS & ACTIVITIES

- Sigma Xi, Inductee
- OUR Scholar, Fall 2025 and Spring 2026 (25% acceptance rate)
- NCUR 2026 Presenter

TECHNICAL SKILLS

Languages: Python, MATLAB, Fortran, C/C++, Java, R

ML & Modeling: PyTorch, Deep Learning, Physics-Informed Neural Networks, Neural ODEs, Transfer Learning, Latent Variable Modeling, Dynamical Systems

Computational: MITgcm, HPC/SLURM, Numerical ODE/PDE Methods, Scientific Computing

Tools: Linux, Git, Docker, React/Node.js, NumPy, scikit-learn